

The Sombor deletion correction term: extremal structure, exact coincidence classification, and the information content of edge decks, with applications to chemistry

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Draft v1 · 2026-09-22 · a companion paper treats the general-kernel deletion framework and the reconstruction theorem

Abstract. For the Sombor index $SO(G) = \sum_{uv \in E} \sqrt{d_u^2 + d_v^2}$, the sum over all edge deletions satisfies $\sum_e SO(G - e) = (m - 1) SO(G) - C(G)$, where the correction term C is itself an edge-additive degree index with an irrational kernel φ . We develop the theory of C : a closed form as a linear functional of edge-type counts, sharp two-sided bounds against Zagreb-type polynomials, an extremal theorem (stars maximise C , with equality characterisation), closed forms for regular graphs and products, and exact certificates that C is not expressible through (n, m, M_1, M_2, F, SO) , the degree sequence, or small spectra. The per-edge drops $\Delta_e = SO(G) - SO(G - e)$ lie in a fixed multiquadratic field; a normal-form procedure yields an exhaustive exact classification at degree bound 8 – a computational theorem (20,707,830 configurations, 10,500,920 exact values): cross-type exact coincidences form exactly two families, $\sqrt{20}$ and $\sqrt{52}$, the within-type coincidences of the tested range are accounted for by a diagonal-union theorem and telescoping identities, and the all-degree statement is conjectural. At deck level, over all 288,266 graphs on $n \leq 9$ vertices and all 89,402 chemical graphs on 10 vertices, the Sombor deck and the edge-type deck induce the same partition (53,984 classes at $n = 10$); on the chemical dataset ($n \leq 9$) the empirical mutual information between C and the edge-type multiset is 9.618 bits, and C is injective modulo matching components on both tested ranges, but fails to be injective in general: $C(5P_4 \sqcup P_6) = C(G_2) = 20\sqrt{2} - 4\sqrt{5}$ for an explicit 8-vertex graph G_2 . On the 18 octane isomers C separates 16 structures and raises a boiling-point regression from $R^2 = 0.559$ (SO alone) to 0.814 (with C); we read this small sample as exploratory. Every identity and inequality is proved or exhaustively verified; the algebraic and extremal core is machine-checked in Lean 4 (status table, Section 8).

1. Introduction

The Sombor index of a graph G , $SO(G) = \sum_{uv \in E} \sqrt{d_u^2 + d_v^2}$ [1], belongs to the family of edge-additive degree indices $T_{f(G)} = \sum_{uv \in E} f(d_u, d_v)$ with symmetric kernel f , which includes the second Zagreb index M_2 [2], the Randić index [3], the ABC index [4] and the Sombor variants surveyed in [5]. The effect of deleting a single edge on Sombor-type indices is known edge-by-edge [6], [7]. Summed over all edges, the deletions obey a closed identity with a single correction term (Proposition 1); the general-kernel framework, its use for reconstruction from the edge deck $\mathcal{D}(G) = \{\{G - e : e \in E\}\}$, and the degenerate-class analysis are the subject of a companion paper. All reconstructibility statements in both papers concern invariants (degree sequences, edge-type counts, edge-additive indices), not graphs; neither paper addresses the edge reconstruction conjecture itself. This paper concerns the correction term itself and the information carried by Sombor-type data at deck level:

- **Structure of C** (Sections 2–3): closed form, sandwiches against Zagreb-type polynomials, extremal theory, non-expressibility certificates.

- **Exact algebra of the drops Δ_e** (Section 4): a complete classification of when two edge deletions produce the same Sombor drop, up to degree 8.
- **Deck information** (Sections 5–7): partition identities, residual entropy, injectivity of C and its exact failure at larger orders, and the octane isomers.

All computations run over exhaustively enumerated datasets: all 288,266 graphs with $n \leq 9$ vertices (273,193 connected; 14,598 chemical, i.e. connected with $\Delta \leq 4$, matching OEIS A000088/A001349/A121941 [8]; canonical labelling by nauty [9]), and all 89,402 chemical graphs with $n = 10$. Equalities between irrational values are decided in exact radical arithmetic (squarefree normal forms); floating-point quantisation is used only to bucket candidates. Section 8 lists the evidence status of every claim in one table.

2. The correction term

Throughout, $n = |V|$, $m = |E|$, d_v is the degree of v , $\Delta = \max_v d_v$, $N(u)$ the neighbourhood of u ; n_{ab} denotes the number of edges with (unordered) degree pair $\{a, b\}$ (the vector (n_{ab}) is the joint edge-degree distribution). Write

$$c(a, b) = \sqrt{a^2 + b^2} - \sqrt{(a-1)^2 + b^2} = \frac{2a-1}{\sqrt{a^2 + b^2} + \sqrt{(a-1)^2 + b^2}}$$

for the Sombor kernel drop.

Proposition 1. Deletion identity. For every graph with $m \geq 1$,

$$\sum_{e \in E} \text{SO}(G - e) = (m-1) \text{SO}(G) - C(G), \quad C(G) = \sum_{u \in V} (d_u - 1) \sum_{x \in N(u)} c(d_u, d_x).$$

More generally, for every symmetric kernel f , with $c_{f(a,b)} = f(a, b) - f(a-1, b)$ and C_f defined analogously,

$$\sum_{e \in E} T_f(G - e) = (m-1) T_f(G) - C_f(G).$$

Proof. Deleting $e = uv$ changes exactly the terms incident to u or v :

$$T_{f(G)} - T_{f(G-e)} = f(d_u, d_v) + \sum_{x \in N(u) \setminus v} c_f(d_u, d_x) + \sum_{y \in N(v) \setminus u} c_f(d_v, d_y).$$

In the sum over all e , each ordered pair (u, x) with $x \in N(u)$ contributes $c_{f(d_u, d_x)}$ once for each deleted edge at u other than ux , i.e. $d_u - 1$ times, and each $f(d_u, d_v)$ contributes once; hence $\sum_e (T_{f(G)} - T_{f(G-e)}) = T_f + C_f$. $\square$

Corollary 2. $\sum_e \text{SO}(G - e) = (m-1) \text{SO}(G)$ if and only if $C(G) = 0$, if and only if G is a matching. Among the 288,249 graphs with $n \leq 9$, $m \geq 2$, the uncorrected identity holds on exactly the 12 matchings and fails on the other 288,237 (maximum relative deviation $36.75\% = 1 - \sqrt{\frac{2}{5}}$, at P_3).

Remark. On P_3 : $\text{SO} = 2\sqrt{5}$, each card is $K_2 \sqcup K_1$ with $\text{SO} = \sqrt{2}$, so $\sum_e \text{SO}(P_3 - e) = 2\sqrt{2} = 1 \cdot \text{SO}(P_3) - C(P_3)$, and $C(P_3) = 2\sqrt{5} - 2\sqrt{2} = 2\varphi(1, 2) \approx 1.644$.

Proposition 3. Closed form. C is edge-additive: with $\varphi(a, b) = (a-1)c(a, b) + (b-1)c(b, a)$,

$$C(G) = \sum_{uv \in E} \varphi(d_u, d_v) = \sum_{a \leq b} n_{ab}(G) w(a, b), \quad w(a, b) = \begin{cases} \varphi(a, b) & \text{if } a \neq b \\ 2(a-1)c(a, a) & \text{if } a = b \end{cases}$$

In particular C is a linear functional of the edge-type count vector, as M_2 is, but with an irrational kernel.

Proposition 4. Elementary evaluations. (i) $C(G) = 0$ iff G is a matching, so $C_{\min}(n, m) = 0$ iff $m \leq \frac{n}{2}$. (ii) If G is r -regular on n vertices, then

$$C = nr(r-1)\left(r\sqrt{2} - \sqrt{2r^2 - 2r + 1}\right).$$

(iii) For Cartesian, tensor and strong products, C has explicit double-sum formulas in the factor degrees.

(iv) C is recoverable from the edge deck whenever $\text{SO}(G)$ is: $C = (m-1)\text{SO}(G) - \sum_e \text{SO}(G-e)$.

3. Inequalities and extremal theory

Write $\Psi_{uv} = (d_u - 1)(2d_u - 1) + (d_v - 1)(2d_v - 1)$ and $\Psi(G) = \sum_{uv \in E} \frac{\Psi_{uv}}{2d_u + 2d_v - 1}$; $M_1 = \sum_v d_v^2$, $F = \sum_v d_v^3$, $Z = 2F - 3M_1 + 2m = \sum_v d_v(d_v - 1)(2d_v - 1)$. Here $\Psi(G)$ is what C becomes when each kernel value $c(a, b)$ is replaced by its rational leading term $\frac{2a-1}{2a+2b-1}$, and Z is the vertex-degree polynomial obtained by clearing denominators; they link the irrational kernel φ to classical polynomial data.

Proposition 5. Sharp sandwiches. For all $a, b \geq 1$,

$$1 \leq c(a, b) \cdot \frac{2a + 2b - 1}{2a - 1} \leq \sqrt{2},$$

asymptotically sharp at both ends ($a = 1, b \rightarrow \infty$; $a = b \rightarrow \infty$). Consequently

$$\Psi(G) \leq C(G) \leq \sqrt{2}\Psi(G), \quad \frac{Z}{4\Delta - 1} \leq C(G) \leq \left(\frac{\sqrt{2}}{3}\right)Z.$$

Both chains have zero violations on all 288,266 graphs; empirically $\frac{C}{\Psi(G)} \in [1.1233, 1.4127]$ (mean 1.3794 over the 273,191 connected graphs with $m \geq 2$). Both inequalities are strict for every graph: the kernel bounds are strict at all finite $a, b \geq 1$, and summing positive terms preserves strictness. The constants are approached asymptotically – along stars ($\frac{C}{\Psi}$ tends to 1) and along regular graphs of unbounded degree ($\frac{C}{\Psi}$ tends to $\sqrt{2}$; at K_9 the ratio is $0.9989\sqrt{2}$).

Proof. For $a, b \geq 1$ write $c(a, b) = \frac{2a-1}{D}$ with $D = \sqrt{a^2 + b^2} + \sqrt{(a-1)^2 + b^2}$. From $\sqrt{x^2 + y^2} \leq x + y$ (for $x, y \geq 0$) we get $D \leq 2a + 2b - 1$, and from $\sqrt{x^2 + y^2} \geq \frac{x+y}{\sqrt{2}}$ we get $D \geq \frac{2a+2b-1}{\sqrt{2}}$; hence $\frac{2a-1}{2a+2b-1} \leq c(a, b) \leq \sqrt{2} \frac{2a-1}{2a+2b-1}$, which is the kernel bound. Multiplying by $d_u + d_v - 2$ and summing over edges gives $\Psi(G) \leq C \leq \sqrt{2}\Psi(G)$; the Z -chain follows by monotonicity of $\frac{1}{t}$ from $3 \leq 2d_u + 2d_v - 1 \leq 4\Delta - 1$. $\square$

Theorem 6. Star maximisation. For every graph with $m \geq 2$ edges,

$$C(G) \leq m(m-1)c(m, 1) = C(K_{1,m}),$$

with equality iff $G \cong K_{1,m}$ plus isolated vertices.

Proof. (1) *Monotonicity.* From the rationalised form, c is strictly increasing in its first and strictly decreasing in its second argument. Since $d_v \geq 1$ and $d_u \leq m$ for every edge, $c(d_u, d_v) \leq c(d_u, 1) \leq c(m, 1)$, and symmetrically with u, v exchanged. (2) *Degree-sum bound.* Every edge satisfies $d_u + d_v \leq m + 1$: vertex v is incident only to uv and to at most $m - d_u$ further edges (those avoiding u). (3) *Zagreb bound.* $M_1 = \sum_{uv \in E} (d_u + d_v)$, because a vertex w contributes d_w to each of its d_w incident edges; hence $M_1 \leq m(m+1)$ by (2), and

$$C = \sum_{uv \in E} [(d_u - 1)c(d_u, d_v) + (d_v - 1)c(d_v, d_u)] \leq c(m, 1) \sum_{uv \in E} (d_u + d_v - 2) = (M_1 - 2m)c(m, 1) \leq m(m-1)c(m, 1).$$

Equality. Equality in (3) forces $d_u + d_v = m + 1$ on every edge. Equality in (1) forces, for each edge and each endpoint, either $d_u = 1$ (the term $(d_u - 1)c$ vanishes) or $c(d_u, d_v) = c(m, 1)$, which by strict monotonicity means $(d_u, d_v) = (m, 1)$; with $d_u + d_v = m + 1$ and $m \geq 2$ this forces every edge to join a degree- m vertex

to a degree-1 vertex. A degree- m vertex is incident with all m edges, hence unique; its component is $K_{1,m}$ and every other vertex is isolated. The converse is immediate. $\square$

The three upper estimates combine into

$$C \leq \min \left\{ \sqrt{2}\Psi(G), \sum_u d_u(d_u - 1)c(d_u, 1), (M_1 - 2m)c(m, 1) \right\}$$

. Measured tightness (mean of C over bound, full dataset): 0.975, 0.697, 0.176 for the three upper bounds; 0.725 ($\Psi(G)$) and 0.550 ($\frac{Z}{4\Delta-1}$) on the lower side. The reverse degree-sequence inequality $\sum_u d_u(d_u - 1)c(d_u, 1) \leq C$ fails on 288,170 graphs. Convexity of $z(d) = d(d - 1)(2d - 1)$ yields an (n, m) -only bound $C \leq \left(\frac{\sqrt{2}}{3}\right)(kz(n - 1) + z(r))$ for $2m = k(n - 1) + r, 0 \leq r < n - 1$; over connected classes its ratio to the true maximum ranges in $[0.061, 0.419]$ and is never tight (Section 9).

Proposition 7. Non-expressibility. C is not a function of any of the following data (exact witness pairs in Appendix D): (n, m, M_1, M_2) ; (n, m, M_1, M_2, F) ; the degree sequence; (n, m, SO) ; the Laplacian spectrum; the adjacency spectrum; the Kirchhoff index. In the (n, m, SO) pair, $\text{SO} = 13\sqrt{2} + 30$ exactly on both graphs while C differs (31.5530 vs. 31.7902); the irrational kernel φ is finer than polynomial degree data.

4. Exact coincidences of the edge-deletion drops

For $e = uv$ the drop $\Delta_e = \text{SO}(G) - \text{SO}(G - e)$ depends only on the local configuration $(a, b, A, B) = (d_u, d_v, (d_x)_{x \in N(u) \setminus v}, (d_y)_{y \in N(v) \setminus u})$:

$$\Delta(a, b, A, B) = \sqrt{a^2 + b^2} + \sum_{x \in A} c(a, x) + \sum_{y \in B} c(b, y).$$

We ask when two configurations give the same value – in particular, when the value fails to determine the edge type $\{a, b\}$ (“cross-type” coincidences).

Normal form. Every radical occurring is $\sqrt{p^2 + q^2}$ with $p, q \leq D$; write $\sqrt{p^2 + q^2} = g\sqrt{s}$ with s square-free. The numbers $\sqrt{s}$, s squarefree, are linearly independent over $\mathbb{Q}$ (multiquadratic extensions; e.g. [10]), so each value $\Delta(a, b, A, B)$ has a unique integer coordinate vector in the basis $\{\sqrt{s} : s \leq 2D^2, s \text{ squarefree}\}$, and equality of two values is decided by comparing coordinates – exactly, without precision thresholds. (Comparing general sums of square roots is a long-standing open problem [11]; bounded degree in a fixed tower is not affected.)

Theorem 8. Classification at degree bound 8; computational. For $D = 8$: among the 20,707,830 unordered configurations (a, b, A, B) with $a, b \leq D$, $|A| = a - 1$, $|B| = b - 1$, entries at most D , the normal form takes 10,500,920 distinct values, and the cross-type coincidences form exactly two families:

$$\Delta(2, 2, [1], [1]) = \Delta(1, 3, [], [1, 6]) = 2\sqrt{5},$$

$$\Delta(3, 3, [2, 3], [2, 2]) = \Delta(2, 4, [1], [3, 4, 6]) = 2\sqrt{13}.$$

For $D = 7$ (1,473,186 configurations, 762,544 values) the same two families are the only ones. Within-type coincidences are abundant (2,402,712 values at $D = 8$); see Propositions 9 and 10.

The computation buckets configuration values as floats at 10^{-6} , then verifies exact coordinates inside every bucket holding more than one configuration or type (runtime 85 s at $D = 8$). Floats play no role in the decision: equal values necessarily share a bucket, so bucketing can only propose candidates, and every proposal is confirmed or refuted by exact coordinate comparison. The basis includes $s = 1$, so rational parts

are tracked; the reduction $\sqrt{p^2 + q^2} = g\sqrt{s}$ is integer squarefree factorisation. The statement for general D is open (Conjecture 11).

quantity	$D = 7$	$D = 8$
configurations (unordered)	1,473,186	20,707,830
distinct exact Δ values	762,544	10,500,920
cross-type coincidence families	2	2
values with within-type coincidences	182,924	2,402,712
float buckets / flagged	—	5,597,558 / 3,346,367
runtime	minutes	85 s

On the $n \leq 9$ dataset, the 119,794 occurring configurations take 85,909 distinct values. A quantised scan had flagged a third candidate near 14.78226, from configurations $(2, 8; (5), (2, 3, 3, 4, 4, 5, 5))$ and $(4, 7; (4, 5, 6), (3, 3, 3, 4, 6, 7))$; their normal forms differ, and the true difference is 4.2389×10^{-10} .

Proposition 9. Diagonal union theorem. For $a \geq 0$ and $|A| = |B| = a - 1$:

$$\Delta(a, a, A, B) = a\sqrt{2} + \sum_{x \in A \cup B} c(a, x),$$

where $A \cup B$ is the multiset union: the value depends only on the union. The map (union $\mapsto$ | value) is injective for every $a \leq 8$ (exhaustive; counts in Appendix F). A general proof would follow from $\mathbb{Q}$ -independence of $\{c(a, x)\}_x$, which is open.

Proposition 10. Telescoping identities. If $a_1^2 + x^2 = (a_2 - 1)^2 + y^2$, then $c(a_1, x) + c(a_2, y) = \sqrt{a_2^2 + y^2} - \sqrt{(a_1 - 1)^2 + x^2}$; if $(a_1 - 1)^2 + x^2 = a_2^2 + y^2$, then $c(a_1, x) + c(a_2, y) = \sqrt{a_1^2 + x^2} - \sqrt{(a_2 - 1)^2 + y^2}$. Distinct telescoping paths collapsing to the same binomial $\sqrt{P} - \sqrt{Q}$ account for all within-type coincidences observed in the tested range, e.g.

$$c(2, 2) + c(3, 2) = \sqrt{13} - \sqrt{5} = c(2, 3) + c(3, 1).$$

Conjecture 11. For every D , the only cross-type coincidences of Δ are the $\sqrt{20}$ and $\sqrt{52}$ families.

Consequences. Δ_e is not a function of the edge type: at $n \leq 9$ the value $2\sqrt{5}$ is attained by $(2, 2)$ -edges (55 configurations) and by $(1, 3)$ -edges (68), and $2\sqrt{13}$ by $(3, 3)$ -edges (410, ordered variants aggregated) and by $(2, 4)$ -edges (20). Both families need a neighbour of degree at least 6 in one of the two configurations, so neither occurs among chemical graphs: at $n = 10$ chemical, the 419 occurring configuration values contain no cross-type coincidence. As multisets, however, $\{\Delta_e\}$ and $\{\text{SO}(G - e)\}$ induce the same partition for $m \geq 4$ (they differ by the shift $\text{SO}(G)$, which the deck recovers there); the single discrepancy at $m = 3$ is C_3 vs. $K_{1,3}$, separated by the Δ -multiset but not by the SO -multiset.

5. The deck information atlas

For an invariant I , the deck signature is the sorted multiset $S_{I(G)} = \text{sort } \{\{I(G - e) : e \in E\}\}$; signatures partition each dataset into classes. We report class counts, the number of graphs sharing a class with a non-isomorphic graph, residual entropy $H(G \text{ mid } S_I) = \sum_C \frac{|C|}{N} \log_2 |C|$ under the uniform prior, and *determination*: whether a target quantity is constant on every class. These measurements are in the discriminating-power tradition (e.g. [12]); the present tables are exhaustive over the stated datasets.

signature	classes	coll. graphs	largest	resid. bits
adj_charpoly	14,598	0	1	0.0000
adj_spec	14,598	0	1	0.0000
lap_spec	14,598	0	1	0.0000
q_spec	14,598	0	1	0.0000
match_poly	14,597	2	2	0.0001
energy	14,597	2	2	0.0001
spectral_radius	14,597	2	2	0.0001
dist_dist	14,039	1,028	6	0.0822
wiener	13,882	1,313	6	0.1055
ecc_seq	13,585	1,832	7	0.1515
sombor	11,112	5,557	20	0.6088
edge_types	11,112	5,557	20	0.6088
randic	11,112	5,557	20	0.6088
abc	11,107	5,564	20	0.6097
taus	7,137	9,142	87	1.9276
deg_seq	1,830	14,051	174	4.2150
m1	1,560	14,165	174	4.4128
chrom	152	14,569	2,297	9.2866
m	19	14,595	3,075	10.8632

Full 48-row table: Appendix A. The five spectral signatures (adj_charpoly, adj_spec, lap_charpoly, lap_spec, q_spec) have zero collisions on all 273,193 connected graphs with $n \leq 9$; edge-reconstructibility of the characteristic polynomial is classical [13]. The graph’s own adjacency spectrum is strictly weaker than the card-spectrum multiset: 1,674 chemical graphs share their own spectrum with a non-isomorphic chemical graph.

Proposition 12. Partition identity. The Sombor-deck signature and the edge-type-deck signature induce the same partition on each of: chemical, chemical planar, chemical with $m \geq 4$, connected, and all graphs at $n \leq 9$; and on all 89,402 chemical graphs at $n = 10$, where both partitions have 53,984 classes.

The Sombor deck therefore carries exactly the card-edge-type information. What it loses relative to the full deck:

	EEj_	EQj0
structure	$K_{3,3}$ minus two independent edges	two triangles joined by a bridge
n, m ; degree sequence	6, 7; 3, 3, 2, 2, 2, 2	6, 7; 3, 3, 2, 2, 2, 2
$\text{SO}(G)$	$7\sqrt{2} + 4\sqrt{13}$	$7\sqrt{2} + 4\sqrt{13}$
$\{\text{SO}(G - e)\}$ multiset	16.9706, 18.7096 ($\times 4$), 20.6067 ($\times 2$)	identical
τ / triangles / W / χ	15 / 0 / 25 / 2	9 / 2 / 27 / 3

The reconstructibility boundary: the 13 edge-deck collision classes at $n \leq 9$ (Appendix B) all have $m \leq 3$ – the edgeless class, the pairs $2K_2 \sqcup (n-4)K_1$ vs. $P_3 \sqcup (n-3)K_1$, and $K_{1,3} \sqcup (n-4)K_1$ vs. $C_3 \sqcup (n-3)K_1$; all 288,208 graphs with $m \geq 4$ have distinct edge decks, as do all 89,402 chemical graphs at $n = 10$. This agrees with the computations of [14] for all graphs to $n \leq 11$; background on the edge reconstruction

conjecture: [15]. Degree sequences are deck-recoverable for $m \geq 4$ (classically [16]; our implementation fails on exactly 20 graphs at $n \leq 9$, all with $m \leq 3$, and on none at $n = 10$ chemical); the companion paper shows that the full edge-type count vector is deck-recoverable when $2\Delta \leq m$. The vertex deck is uniformly stronger for degree-based signatures (Sombor residual entropy 0.037 vs. 0.609 bits on chemical graphs) and has a single collision class, K_2 vs. $2K_1$.

check	$n \leq 9$ (all 288,266)	$n = 10$ chemical (89,402)
edge-deck collisions	13 classes, all $m \leq 3$; none at $m \geq 4$	none
deck $\rightarrow$ degree sequence	20 failures, all $m \leq 3$	0 failures
Sombor deck == edge-type deck	same partition (5 datasets)	same partition, 53,984 classes
cross-type Δ families occurring	2 ($\sqrt{20}$, $\sqrt{52}$)	0 (degrees ≤ 4)
C collisions mod matchings	0; min gap 3.2×10^{-8}	0; min gap 6.4×10^{-5}

6. Information content and injectivity of C

By Proposition 4(iv), C is a deck invariant exactly when $\text{SO}(G)$ is; on the class $m \geq 4$, $2\Delta \leq m$ this holds unconditionally (companion paper). To measure C as a scalar summary, let et be the edge-type multiset of G , viewed as a random variable under the uniform distribution on the dataset, and let $I(C; \text{et})$ and $H(\text{et} \mid C)$ be the corresponding mutual information and conditional entropy. On the chemical dataset ($n \leq 9$), C attains the highest mutual information among the tested scalars:

scalar X	distinct values	$I(X; \text{et})$ bits	$H(\text{et} \mid X)$ bits	mean level set
C	1,829	9.618	0.0001	34.9
SO	1,823	9.614	0.0048	35.0
ABC	1,615	9.530	0.0889	35.4
Randić	1,076	8.560	1.0587	75.8
M_2	213	7.258	2.3604	109.3
τ	2,144	5.842	3.7762	37.6
degree seq.	300	6.378	3.2402	295.1
M_1	64	5.186	4.4326	473.7
W	111	3.119	6.4993	448.9
m	19	2.970	6.6482	2181.4

The edge-type multiset of the chemical $n \leq 9$ dataset has entropy 9.618 bits over 1,830 distinct multisets (uniform distribution over the 14,598 graphs); the single identification made by C is $C(K_1) = C(K_2) = 0$, a matching shift. Since $\varphi(1, 1) = 0$, adding or removing K_2 components never changes C ; call two edge-type multisets *matching-equivalent* if they differ by $(1, 1)$ -cells only. Injectivity is stated modulo this equivalence.

Proposition 13. Injectivity modulo matchings, finite range. C is injective on matching-equivalence classes of edge-type multisets among: all graphs with $n \leq 9$ (55,087 distinct exact values; the 733 raw collision classes all reduce to matching shifts; minimum gap between distinct values 3.206×10^{-8} , witness pair in Appendix D), and all chemical graphs with $n = 10$ (2,223 values; minimum gap 6.4×10^{-5}).

Injectivity fails without the range restriction:

Example 14. Exact collision. Let $G_1 = 5P_4 \sqcup P_6$ (26 vertices, 20 edges; edge-type multiset $12 \times (1, 2) + 8 \times (2, 2)$) and let G_2 have vertices $a_1, a_2, a_3, a_4, b_1, b_2, c_1, c_2$ and edges $a_1a_2, a_2a_3, a_1a_3, a_4b_1, a_4b_2, a_1b_1, a_2b_2, a_4c_1, a_3c_2$ (degree sequence $3, 3, 3, 3, 2, 2, 1, 1$; multiset $3 \times (3, 3) + 4 \times (2, 3) + 2 \times (1, 3)$). Then

$$C(G_1) = 12\varphi(1, 2) + 8\varphi(2, 2) = 2\varphi(1, 3) + 4\varphi(2, 3) + 3\varphi(3, 3) = C(G_2) = 20\sqrt{2} - 4\sqrt{5} \approx 19.3400,$$

an exact identity in $\mathbb{Q}(\sqrt{2}, \sqrt{5}, \sqrt{10}, \sqrt{13})$; neither graph is a matching perturbation of the other. Both satisfy $M_1 - 2m = 28$.

Collisions are lattice points. Writing edge-type count vectors N , $C(N_1) = C(N_2)$ if and only if $N_1 - N_2$ lies in the integer relation lattice $\Lambda_D = \left\{ v : \sum_{x,y} v_{xy} \varphi(x, y) = 0 \right\}$ of the φ -weights in the squarefree basis: $\dim \Lambda_D = 3, 4, 8, 10, 15$ for $D = 4, 5, 6, 7, 8$ (9 cells and basis radicals $1, \sqrt{2}, \sqrt{5}, \sqrt{10}, \sqrt{13}, \sqrt{17}$ at $D = 4$; 35 cells, rank 20, at $D = 8$). Abstract relations are plentiful, so injectivity in finite ranges comes from realisability constraints. One such constraint is linear: for $D \leq 6$ the weight vector $w = (x + y - 2)_{xy}$ is orthogonal to Λ_D (verified by exact rational linear algebra), so every collision whose cells have degrees at most 6 conserves $M_1 - 2m$ — as in Example 14, where both graphs have $M_1 - 2m = 28$. This conservation is not a general law: it fails from degree 7 on.

Example 15. Conservation fails. Let $G_1 = 630P_3 \sqcup 280C_3$ and $G_2 = 70K_{1,3} \sqcup 35K_{1,4} \sqcup 21K_{1,5} \sqcup 14K_{1,6} \sqcup 10K_{1,7}$. Then

$$\begin{aligned} C(G_1) &= 1260\varphi(1, 2) + 840\varphi(2, 2) \\ &= 210\varphi(1, 3) + 140\varphi(1, 4) + 105\varphi(1, 5) + 84\varphi(1, 6) + 70\varphi(1, 7) = C(G_2) \\ &= 2100\sqrt{2} - 420\sqrt{5}, \end{aligned}$$

an exact identity: the right-hand side telescopes to $420(\sqrt{50} - \sqrt{5})$ because $\sqrt{50} = 5\sqrt{2}$, which equals the left-hand side. But $M_1 - 2m$ equals 2940 on G_1 and 2100 on G_2 . Taking t disjoint copies of both graphs gives an infinite family of exact collisions with unbounded $M_1 - 2m$ discrepancy.

A decision theorem — collisions as lattice points realisable on both sides, and which linear statistics realisable collisions must conserve — and a threshold $n_0(\Delta)$ below which injectivity holds ($n_0 \leq 26$ by Example 14) are open (Section 9).

7. Octane isomers

On the standard test set of the 18 octane isomers (full data in Appendix C; boiling points from the classic QSPR benchmark of [17], cross-checked against NIST WebBook within stated uncertainties):

index	distinct /18	Pearson (BP)	Spearman (BP)	OLS R^2
C	16	−0.760	−0.837	0.577
SO	16	−0.748	−0.837	0.559
ABC	15	−0.862	−0.928	
Randić	16	+0.819	+0.877	
M_1	6	−0.718	−0.739	0.516
W [18]	16	+0.537	+0.488	
$SO + C$				0.814
$SO + M_1 + C + W$				0.959

With $n = 18$ we read these regressions as exploratory. On the octane set, correlation 0.9997, so C is nearly indistinguishable from SO as a single descriptor; the pair $SO + C$ has VIF approximately 1713 (extreme

collinearity), hence the joint model is predictive but its coefficients admit no mechanistic reading. The component of C orthogonal to SO is small (residual standard deviation 2.4 percent of C 's), yet it correlates at minus 0.76 with the SO-residual of boiling point: the increment of SO + C over SO (R^2 0.559 $\rightarrow$ 0.814; adjusted 0.532 $\rightarrow$ 0.789; $F(1, 15) = 20.5$, $p = 4 \times 10^{-4}$; LOO RMSE 4.58 $\rightarrow$ 3.01) is a genuine second-order correction, not a quadratic artefact (SO + SO² reaches only 0.605) nor a branching surrogate ($n_3 + n_4$ reaches 0.539). Against ABC, however, the increment disappears (ABC + C : 0.751 vs. ABC alone 0.743).

Across five properties (boiling point; entropy; acentric factor; enthalpy of vaporisation; standard enthalpy of vaporisation, all from the same benchmark), C correlates strongly throughout ($|r| = 0.76\text{--}0.96$) but is never the best single index: M_1 or M_2 win on entropy and acentric factor, Randić on both vaporisation enthalpies, and ABC on boiling point. SO + C improves on the best single index for BP, HVAP and DHVAP but not for entropy or acentric factor (compared at equal parameter count, i.e. adjusted R^2). C ties exactly on the two isomer pairs with identical edge-type multisets (3- and 4-methylheptane; 3,4-dimethylhexane and 3-ethyl-2-methylpentane); the ties are shared with SO and are structural, not numerical.

Within fixed (n, m) over the chemical dataset, the mean Spearman correlation of C is +0.967 with degree variance, +0.742 with leaf count, -0.567 with spanning-tree count, +0.439 with the Kirchhoff index, -0.282 with branching-vertex count and +0.026 with W : C measures deviation of the degree distribution from regularity, not molecular shape.

8. Verification and data availability

claim	status
Deletion identity, general kernel (Prop. 1)	proved; Lean <code>deletion_identity</code> (unconditional) with local formula <code>Tf_delete_singleton</code> ; 12 indices exhaustively verified, 0 failures
Naive identity iff matching (Cor. 2)	proved; exhaustive (288,249 graphs; the 12 exceptions are exactly the matchings)
Closed form (Prop. 3)	algebraic regrouping; exhaustive over 55,855 edge-type multisets (max deviation 3.1×10^{-13})
Zeros; regular value (Prop. 4 i–ii)	Lean <code>C_eq_zero_iff</code> , <code>C_of_regular</code>
Sandwiches (Prop. 5)	Lean <code>psiSum_le_C</code> , <code>C_le_sqrt2_psiSum</code> , <code>zagreb_sandwich</code> ; 0 violations on 288,266 graphs
Star maximisation and equality (Thm. 6)	Lean <code>C_le_star</code> , <code>C_eq_star_iff</code> ; 0 violations on 288,257 graphs; all 28 equality cases are stars
Non-expressibility (Prop. 7)	witness pairs re-verified in exact radical / integer arithmetic; the Kirchhoff pair P_3 vs. the diamond K_4 minus an edge is connected (Appendix D)
Classification (Thm. 8; computational)	exhaustive verified computation at $D \leq 8$; the two family identities Lean-certified (<code>Delta_sqrt20_*</code> , <code>Delta_sqrt52_*</code> , <code>sqrt20_ne_sqrt52</code>); general D open (Conj. 11)
Diagonal union; telescoping (Props. 9–10)	Lean <code>Delta_diagonal_union</code> , <code>telescoping_one</code> , <code>telescoping_two</code> ; union injectivity exhaustive for $a \leq 8$
Atlas tables; partition identity (Prop. 12)	exhaustive; enumeration validated against OEIS and the graph atlas; partition comparisons persisted per dataset
Injectivity of C (Prop. 13)	exact normal-form coordinates; minimum-gap witness in Appendix D

claim	status
Collision pair (Ex. 14)	explicit construction; radical-arithmetic difference exactly 0
$M_1 - 2m$ conservation	exact rational linear algebra: $w \perp \Lambda_D$ for $D \leq 6$; violated at $D \geq 7$ (Ex. 15, exact construction)
Octane numbers (Section 7)	recomputed from molecular graphs; exploratory ($n = 18$); property data from the QSPR benchmark [17], cross-checked against NIST WebBook

The Lean 4 [19] + Mathlib [20] library accompanying this work builds with 97 theorems and no sorry; the inventory and the exact boundary of the formalisation (identities, mechanisms and certificates are formalised; the classification and the companion paper’s recovery theorem are not) are documented there. The data pipeline reproduces every result file in one pass (`regenerate_all.sh`); enumeration, invariants and headline witnesses pass 96 automated checks (OEIS counts, NetworkX atlas, independent implementations, exact re-verification of every quoted witness), and all inequalities were additionally stress-tested on structured graph families up to $n = 300$.

Code and data availability. The complete source package — the Lean 4 formalisation, the Python verification pipeline, every result file, and the compiled manuscript — is publicly available at https://github.com/JGong-CatenaryGong/sombor_deletion_correction a DOI-stamped archival snapshot will accompany the final version.

9. Open problems

- **Classification beyond degree 8.** Conjecture 11; parameterise the two families; prove diagonal-union injectivity in general ($\mathbb{Q}$ -independence of $\{c(a, x)\}_x$).
- **Minimum of C at fixed (n, m) .** The argmin degree sequence is quasi-regular in all 92 connected (n, m) -classes at $n \leq 9$; conjectured closed form $C_{\min} = (n - k)G(r) + kG(r + 1) + t_{\min}\varphi^*(r)$ for $2m = nr + k$, where $G(d) = d(d - 1)c(d, d)$ is strictly convex on integer points, $\varphi^*(r)$ is the margin of mixed edges, and $t_{\min}$ is the forced number of mixed $r - (r + 1)$ edges; the key step is the binary inequality $\varphi(a, b) \geq h(a) + h(b)$, $h(d) = (d - 1)c(d, d)$, verified on a 400^2 grid with equality iff $a = b$. Corollaries would include: among trees the path minimises and the star maximises C . Also open: a tight (n, m) -upper bound.
- **Collisions of C .** A decision theorem via Λ_D -realisability; which linear statistics are conserved by realisable collisions ($M_1 - 2m$ is forced at $D \leq 6$ but not from degree 7 on, Examples 14–15); the threshold $n_0(\Delta)$; asymptotics as $\dim \Lambda_D$ grows ($3 \rightarrow 15$ from $D \leq 4$ to $D \leq 8$).
- **Formalisation.** Multiquadratic linear independence in usable form, then the normal-form procedure and Theorem 8; formal decks, then the companion paper’s recovery theorem.

Usage of generative AI

The mathematical content of this paper was developed with the assistance of large language models used strictly as research tools. A large reasoning model (Kimi K3) was used for problem framing, hypothesis generation and verification design; a fast small model (deepseek v4.1-flash) for bulk exploratory computation; after each stage a large model was consulted in a fresh session with no shared context for adversarial review.

Every identity, inequality and extremal statement was independently re-derived by the author and, where indicated, formalised in Lean 4; every coincidence, collision, gap and counterexample was re-verified in exact arithmetic, never by floating point alone.

Appendix A — Deck signatures on chemical graphs, full table

signature	classes	coll. graphs	rate	largest	resid. bits
itself	14,598	0	0.0000	1	0.0000
adj_spec	14,598	0	0.0000	1	0.0000
lap_spec	14,598	0	0.0000	1	0.0000
q_spec	14,598	0	0.0000	1	0.0000
adj_charpoly	14,598	0	0.0000	1	0.0000
lap_charpoly	14,598	0	0.0000	1	0.0000
match_poly	14,597	2	0.0001	2	0.0001
energy	14,597	2	0.0001	2	0.0001
spectral_radius	14,597	2	0.0001	2	0.0001
dist_dist	14,039	1,028	0.0704	6	0.0822
wiener	13,882	1,313	0.0899	6	0.1055
ecc_seq	13,585	1,832	0.1255	7	0.1515
edge_types	11,112	5,557	0.3807	20	0.6088
sombor	11,112	5,557	0.3807	20	0.6088
sombor_inv	11,112	5,557	0.3807	20	0.6088
sombor_red	11,112	5,557	0.3807	20	0.6088
randic	11,112	5,557	0.3807	20	0.6088
ga	11,111	5,559	0.3808	20	0.6090
isi	11,111	5,559	0.3808	20	0.6090
abc	11,107	5,564	0.3811	20	0.6097
az	11,107	5,564	0.3811	20	0.6097
sum_conn	11,103	5,569	0.3815	20	0.6106
harmonic	11,101	5,572	0.3817	20	0.6109
m2	10,711	6,122	0.4194	20	0.6826
taus	7,137	9,142	0.6263	87	1.9276
n_cut	2,081	13,799	0.9453	528	4.8113
deg_seq	1,830	14,051	0.9625	174	4.2150
sombor_sq	1,829	14,053	0.9627	174	4.2151
forgotten	1,829	14,053	0.9627	174	4.2151
diam	1,577	13,952	0.9557	313	5.1957
m1	1,560	14,165	0.9703	174	4.4128
n_leaves	583	14,449	0.9898	653	6.6612
comp_sizes	482	14,427	0.9883	1,585	8.1729
n_bridges	451	14,532	0.9955	622	6.6223
n_perfect_match	363	14,443	0.9894	2,778	9.8708
tri	350	14,520	0.9947	529	7.2348
indep	302	14,534	0.9956	796	7.5182
min_deg	179	14,565	0.9977	1,331	8.2561
girth	156	14,558	0.9973	2,138	9.1749
chrom	152	14,569	0.9980	2,297	9.2866
biconnected	113	14,581	0.9988	2,005	9.0377
clique	85	14,586	0.9992	1,927	9.1684

signature	classes	coll. graphs	rate	largest	resid. bits
bipartite	82	14,585	0.9991	2,507	9.7516
max_match	72	14,588	0.9993	2,936	10.0827
n_comp	70	14,591	0.9995	1,711	9.2565
max_deg	61	14,592	0.9996	2,555	9.7867
n	53	14,589	0.9994	2,768	10.2503
forest	45	14,587	0.9992	3,075	10.7646
m	19	14,595	0.9998	3,075	10.8632

Appendix B — The 13 edge-deck collision classes at $n \leq 9$

m	n range	# graphs	members (graph6)
0	1–9	9	@, A?, B?, C?, D??, E???, F????, G?????, H??????
2	4–4	2	CE, CQ
2	5–5	2	D?o, DC0
2	6–6	2	E?B?, E?`?
2	7–7	2	F??E?, F?AA?
2	8–8	2	G???E?, G??CA?
2	9–9	2	H????B?, H???C@?
3	4–4	2	CF, CT
3	5–5	2	D?w, DCc
3	6–6	2	E?B_, E?aG
3	7–7	2	F??F?, F?ACG
3	8–8	2	G???F?, G??CCC
3	9–9	2	H????B_, H???CA@

Appendix C — Octane isomers, full data

molecule	degree seq.	C	SO	ABC	M_1	W	BP
n-octane	2,2,2,2,2,2,1,1	7.5673	18.6143	4.9497	26	84	125.7
2-methylheptane	3,2,2,2,2,1,1,1	10.0784	20.6515	5.1685	28	79	117.6
3-methylheptane	3,2,2,2,2,1,1,1	9.8606	20.5024	5.0591	28	76	118.9
4-methylheptane	3,2,2,2,2,1,1,1	9.8606	20.5024	5.0591	28	75	117.7
3-ethylhexane	3,2,2,2,2,1,1,1	9.6428	20.3533	4.9497	28	72	118.5
2,2-dimethylhexane	4,2,2,2,1,1,1,1	14.7875	24.7344	5.4265	32	71	106.8
2,3-dimethylhexane	3,3,2,2,1,1,1,1	12.1097	22.3995	5.2375	30	70	115.6
2,4-dimethylhexane	3,3,2,2,1,1,1,1	12.3717	22.5396	5.2779	30	71	109.4
2,5-dimethylhexane	3,3,2,2,1,1,1,1	12.5894	22.6886	5.3873	30	74	109.1
3,3-dimethylhexane	4,2,2,2,1,1,1,1	14.4910	24.4910	5.2676	32	67	111.9
3,4-dimethylhexane	3,3,2,2,1,1,1,1	11.8919	22.2504	5.1281	30	68	117.7
2,2,3-trimethylpentane	4,3,2,1,1,1,1,1	16.6471	26.3732	5.4743	34	63	109.8
2,2,4-trimethylpentane	4,3,2,1,1,1,1,1	17.2986	26.7716	5.6453	34	66	99.2
2,3,3-trimethylpentane	4,3,2,1,1,1,1,1	16.5683	26.2790	5.4248	34	62	114.8
2,3,4-trimethylpentane	3,3,3,1,1,1,1,1	14.3588	24.2967	5.4158	32	65	113.5
3-ethyl-2-methylpentane	3,3,2,2,1,1,1,1	11.8919	22.2504	5.1281	30	67	115.6

molecule	degree seq.	C	SO	ABC	M_1	W	BP
3-ethyl-3-methylpentane	4,2,2,2,1,1,1,1	14.1944	24.2477	5.1087	32	64	118.3
2,2,3,3-tetramethylbutane	4,4,1,1,1,1,1,1	21.2360	30.3955	5.8085	38	58	106.5

Appendix D – Certificates and witnesses

Graph identifiers in this appendix are graph6 strings (the character ? encodes six zero bits).

Non-expressibility pairs (Prop. 7):

C not determined by	pair	n, m	shared	C values
(n, m, M_1, M_2)	ECQo / ECp0	6, 5	$M_1 = M_2 = 20$	7.032181 / 5.923591
(n, m, M_1, M_2, F)	F?`b_ / F?qc_	7, 6	deg. seq., $M_1 = M_2 = 24$, $F = 54$	8.216900 / 8.458129
degree sequence	DEg / DQg	5, 4	$(2, 2, 2, 1, 1)$: P_5 vs $K_3 \sqcup K_2$	4.013145 / 3.554155
(n, m, SO)	F?zf0 / FCXjW	7, 10	$\text{SO} = 13\sqrt{2} + 30$ exactly	31.553035 / 31.790213
Laplacian spectrum	ECZo / ECz_	6, 7	characteristic polynomials equal	15.349292 / 14.939222
adjacency spectrum	D?{ / DEo	5, 4	spectra equal	11.529936 / 4.738873
Kirchhoff index	BW / C^	3, 2; 4, 5	Kf = 4 both (eigenvalues 0, 1, 3 resp. 0, 2, 4, 4); both connected	$2\sqrt{5} - 2\sqrt{2}$ / $8\sqrt{13} - 4\sqrt{10} - 4\sqrt{2}$

The Kirchhoff pair replaces an earlier disconnected witness: P_3 and the diamond K_4 minus an edge are both connected, so the standard definition $\text{Kf} = n \sum_{\mu \neq 0} \frac{1}{\mu}$ over the nonzero Laplacian eigenvalues applies without convention; C differs exactly ($2\sqrt{5} - 2\sqrt{2} \approx 1.644$ vs.

$8\sqrt{13} - 4\sqrt{10} - 4\sqrt{2} \approx 10.538$). Adjacency data for every witness pair are included in the data release.

Minimum C -gap pair at $n \leq 9$ (distinct edge-type multisets, connected graphs): H?`EVVv and G?rmv{, with $C = 92.2888805107552$ vs. 92.28888054281828 ; gap 3.206307×10^{-8} .

The two cross-type Δ families (exact identities, Lean-certified):

$$\sqrt{8} + 2(\sqrt{5} - \sqrt{2}) = 2\sqrt{5} = \sqrt{10} + (\sqrt{10} - \sqrt{5}) + (3\sqrt{5} - 2\sqrt{10}),$$

and both $\sqrt{52}$ -configurations simplify to $2\sqrt{13}$. The resolved near-miss pair has distinct normal-form coordinate vectors (difference $4.2389 \times 10^{-10} \neq 0$); the vectors are listed in the data release.

Appendix E – Determination matrix, chemical graphs with $m \geq 4$

deck signature	ds	et	so	dd	taus	tri	wiener	n	m	C	ds_sig
deg_seq	D	D	D	.	.	.	.	D	D	D	D
edge_types	D	D	D	D	.	.	.	D	D	D	D
n_comp	.	.	.	.	.	.	.	.	D	.	.
comp_sizes	.	.	.	.	.	.	.	D	D	.	.
tri	.	.	.	.	.	D	.	.	D	.	.
clique	.	.	.	.	.	.	.	.	D	.	.
chrom	.	.	.	.	.	.	.	.	D	.	.
girth	.	.	.	.	.	.	.	.	D	.	.
n_bridges	.	.	.	.	.	.	.	.	D	.	.
n_cut	.	.	.	.	.	.	.	.	D	.	.
diam	.	.	.	.	.	.	.	.	D	.	.
dist_dist	.	.	.	.	.	.	D	D	D	.	.

deck signature	ds	et	so	dd	taus	tri	wiener	n	m	C	ds_sig
match_poly	D	D	D	D	D	D	D	D	D	D	D
taus	.	.	.	.	D	.	.	D	D	.	.
sombor	D	D	D	D	.	.	.	D	D	D	D
sombor_inv	D	D	D	D	.	.	.	D	D	D	D
randic	D	D	D	D	.	.	.	D	D	D	D
abc	D	D	D	.	.	.	.	D	D	D	D
m1	.	.	.	.	.	.	.	.	D	.	.
m2	D	D	D	.	.	.	.	D	D	D	D
wiener	.	.	.	.	.	.	.	D	D	.	.
adj_spec	D	D	D	D	D	D	D	D	D	D	D
lap_spec	D	D	D	D	D	D	D	D	D	D	D
q_spec	D	D	D	D	D	D	D	D	D	D	D
adj_charpoly	D	D	D	D	D	D	D	D	D	D	D
lap_charpoly	D	D	D	D	D	D	D	D	D	D	D
energy	D	D	D	D	D	D	D	D	D	D	D

D = the deck signature determines the target on every class (14,592 graphs). Targets: ds degree sequence, et edge-type multiset, so $SO(G)$, dd $\{\Delta_e\}$ multiset, taus spanning trees, tri triangles, wiener W , n, m, ds_sig card degree-sequence signature, C correction term.

Appendix F — Diagonal-union counts at $D = 8$

a	unions	values	injective
2	1	1	yes
3	267	267	yes
4	1,653	1,653	yes
5	6,378	6,378	yes
6	19,404	19,404	yes
7	50,377	50,377	yes
8	116,241	116,241	yes

Counts cover unions appearing in collision-flagged buckets; unions in singleton buckets are trivially injective, so injectivity holds on the full diagonal range (Prop. 9). At $D = 7$ the check covered all unions: counts $\binom{2a+4}{2a-2}$, $a = 1..7$.

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